№1

$$|M|(E) = \sup \sum_{i=1}^{\infty} |M(E_i)|$$

1) Пусть $E_i = \emptyset$ при $i > N$:

$$\sup \sum_{i=1}^{N} |M(E_i)| = \sup \sum_{i=1}^{\infty} |M(E_i)| \le |M|(E) \Rightarrow$$

$$\Rightarrow \lambda(E) \le |M|(E)$$

2) $\forall \varepsilon > 0 \ \exists \{A_i\}_{i=1}^{\infty} : \sum_{i=1}^{\infty} |M(A_i)| > |M|(E) - \frac{\varepsilon}{2}$

$$\sum_{i=1}^{\infty} |M(A_i)| - \text{сх.} \Rightarrow \exists N : \sum_{i=N+1}^{\infty} |M(A_i)| < \frac{\varepsilon}{2}$$

Пусть $E_1 = A_1, \ldots, E_N = A_N, E_{N+1} = \bigcup_{i=N+1}^{\infty} A_i$

$$\sum_{i=1}^{N+1} |M(E_i)| = \sum_{i=1}^{N} |M(A_i)| + |M(\bigcup_{i=N+1}^{\infty} A_i)|$$

$$\sum_{i=1}^{N+1} |M(E_i)| \ge \sum_{i=1}^{N} |M(A_i)| = \sum_{i=1}^{\infty} |M(A_i)| - \sum_{i=N+1}^{\infty} |M(A_i)| >$$

$$> (|M|(E) - \frac{\varepsilon}{2}) - \frac{\varepsilon}{2} = |M|(E) - \varepsilon$$

$$\lambda(E) \ge \sum_{i=1}^{N+1} |M(E_i)| > |M|(E) - \varepsilon$$

$$\lambda(E) \ge |M|(E)$$

№2

$$\#(A) = 0 \Rightarrow A = \emptyset \Rightarrow dx(A) = 0$$

$\#$ не имеет разложения отн. dx

№3

$$f \in L^p([0,1]) \Leftrightarrow \int_0^1 |f(x)|^p dx < \infty$$

$$f \in L^{\infty}([0,1]) \Leftrightarrow \sup_{x \in [0,1]} |f(x)| < \infty \text{ п.в.}$$

$$\int_0^1 |f(x)|^p dx \le \int_0^1 M^p dx \le M^p \quad \forall p : 1 \le p < \infty$$

$$\Rightarrow f \in L^p([0,1])$$

$$f = \ln x$$

$$\int_0^1 |\ln x|^p dx = \left| \begin{array}{l} x = e^{-t} \\ dx = -e^{-t}dt \\ x(0) = +\infty \\ x(1) = 0 \end{array} \right| = \int_0^{+\infty} t^p e^{-t} dt = \int_0^1 t^p e^{-t} dt + \int_1^{+\infty} t^p e^{-t} dt$$

$$0 \le t^p e^{-t} \le 1 \text{ при } t \in [0,1]$$

$$t^p \le e^{t/2} \text{ при } t \to +\infty \Rightarrow t^p e^{-t} \le e^{-t/2}$$

$$\int_1^{\infty} t^p e^{-t} dt \le \int_1^{\infty} e^{-t/2} dt = 2e^{-1/2} < \infty$$

$$\Rightarrow f \in \bigcup_{1 \le p < \infty} L^p([0,1]), \text{ но } f \notin L^{\infty}([0,1])$$

$$\Rightarrow L^{\infty}([0,1]) \subset \bigcup_{1 \le p < \infty} L^p([0,1])$$